

Edwin Liu

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EDUCATION

Purdue University, College of Engineering

Bachelor of Science in Mechanical Engineering, Minor in Management, Minor in Chinese

West Lafayette, IN
Aug 2023 – May 2027

PROFESSIONAL EXPERIENCE

GE Appliances, a Haier Company

Advanced Development Engineer

Louisville, KY
Jan 2026 – May 2026

Microwave Test & Validation

- Performed testing and validation on new microwave technology, achieving **96% accuracy** against target specifications.
- Used **Python** and heat transfer principles to develop a power profile with a global engineering team in India.
- Automated microwave testing using **Postman API** and the SmarthQ mobile application.
- Presented technical findings monthly to product managers, influencing design decisions.

Water Quality Sensor Development

- Designed a water-quality sensing subsystem in **Creo Parametric** with a team of mechanical and electrical engineers.
- Designed custom water conductivity sensor to fit within subsystem housing and calibrated it with **Arduino IDE** and **Excel**.
- Filed an invention disclosure for a self-designed, novel feature within a water quality sensing system
- Demonstrated a functional prototype at the Co-op Design Competition, communicating feasibility and performance to engineers.

Eagle Vision and Automation

Automation Engineer

Yorba Linda, CA
May 2025 – Jul 2025

- Automated sandblaster motion using **DirectSOFT** ladder logic programming, reducing setup time by **70%**.
- Programmed **Human-Machine Interface (HMI)** screens using C-more Micro software.
- Programmed a PLC for a helicoil driver with custom 3D-printed components, reducing defects by **33%**.
- Developed an automated tray-removal process to clear conveyor buildup, reducing the need for **2 operators**.
- Created a **SolidWorks** wiring schematic for the MiraAI Vision Robot.

PROJECTS

Heat Sink Optimization Project

Thermal Systems Researcher

West Lafayette, IN
Fall 2026

- Designed an experimental system to evaluate how heat sink geometry and coolant type affect semiconductor heat transfer.
- Developed a PVC flow loop with a machined test plate, pump, heating element, and thermocouples to measure thermal performance.
- Designed housing apparatus for experiment using **Siemens NX**
- Applied heat transfer principles to compare materials, thermal pastes, and coolants to determine an optimal heat sink configuration.

American Society of Mechanical Engineers (ASME)

Small Projects Team Design Leader

West Lafayette, IN
Aug 2023 – Dec 2023

- Designed, built, and tested a racing drone from scratch for the Purdue ASME Competition with a \$150 budget.
- 3D modeled and iteratively prototyped a drone frame in **SolidWorks** to securely house all components.
- Created and analyzed a Bill of Materials to evaluate propellers, motors, flight controllers, and batteries.
- Soldered and assembled electronic components, ensuring reliable connections and full integration of the flight control system.

Tank Gacha Game

JavaScript Programmer

Fullerton, CA
May 2023 – Aug 2023

- Programmed a complete two-player game in **JavaScript** using Greenfoot, including player controls, combat mechanics, and a gacha-based tank and weapon system.

TECHNICAL SKILLS

- Data Analysis:** Excel Modeling, MATLAB, Simulink
- CAD / Design:** Creo Parametric, SolidWorks, Siemens NX, Fusion 360, Abaqus FEA, 3D Printing
- Programming / Automation:** Python, C/C++, MATLAB, PLC Programming (DirectSOFT06), Postman API
- Electrical / Prototyping:** Breadboarding, Soldering, Arduino IDE

EXTRACURRICULARS

- Intramural Basketball Captain, Symphony Orchestra Section Leader, Purdue Chinese Calligraphy Club